

Boundary Conditions :
Dielectric media.

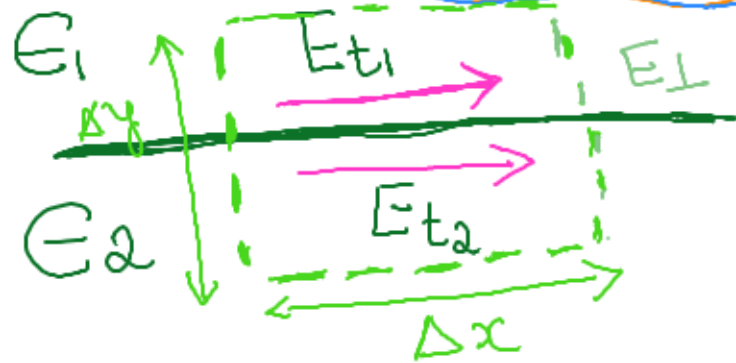
static $\vec{E}$ field around a closed path

$\oint \vec{E} \cdot d\vec{l}$ The line integral of the



$$\oint \vec{E} \cdot d\vec{l} = 0$$

static field

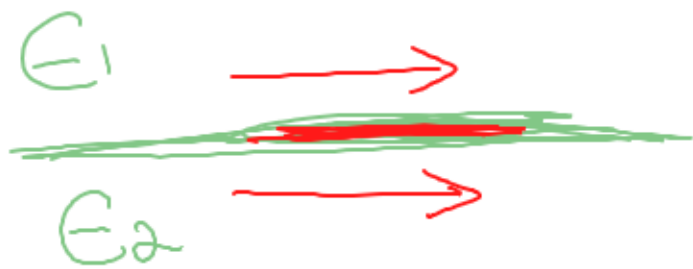


Tangential $\vec{E}$ field
 at the boundary

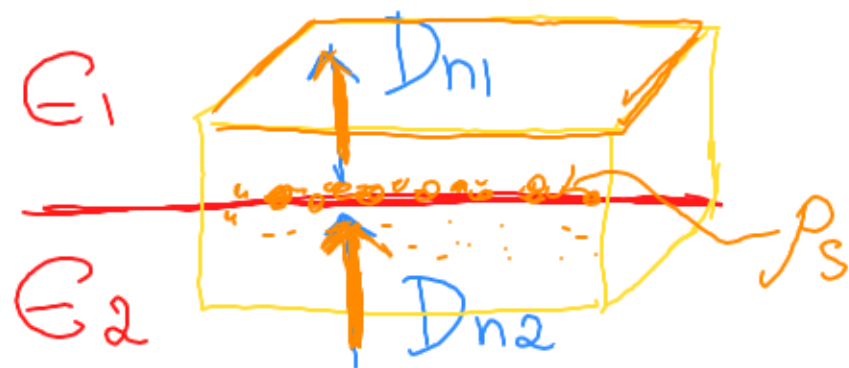
$$E_{t1} \Delta x + \vec{E}_\perp \Delta y - E_{t2} \Delta x - \vec{E}_\perp \Delta y = 0$$

$$E_{t1} \cancel{\Delta x} - E_{t2} \cancel{\Delta x} = 0$$

$$\boxed{E_{t1} - E_{t2} = 0}$$



The tangential $\vec{E}$ field is continuous at the boundary.



$$\oint \vec{D} \cdot d\vec{s} = Q$$

$$\oint \vec{D} \cdot d\vec{s} = P_s A$$

$$\textcircled{D} = \epsilon \textcircled{E}$$

$$\cancel{D_{n1} A} - \cancel{D_{n2} A} = 0$$

$$D_{n1} - D_{n2} = 0$$

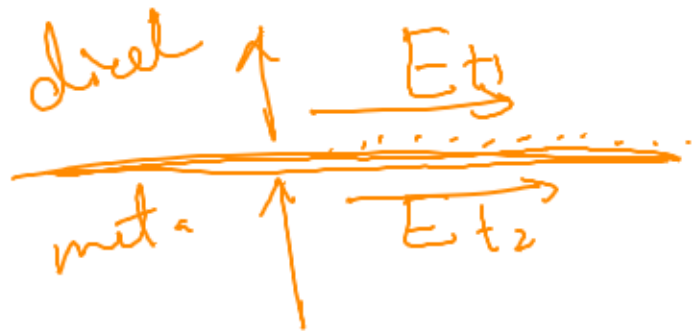
Normal $\vec{D}$ is continuous.

dielectric - dielectric media

$$\vec{E}_{t1} - \vec{E}_{t2} = 0 \quad \text{continuous}$$

$$\vec{D}_{n1} - \vec{D}_{n2} = 0 \quad \text{continuous}$$

dielectric - metal.



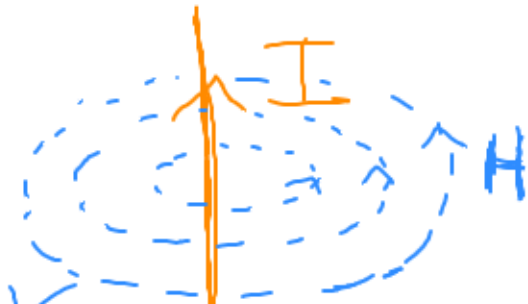
$$\vec{E}_{t1} - \vec{E}_{t2} = 0 \quad \text{continuous}$$

$$\vec{D}_{n1} - \vec{D}_{n2} = \rho_s$$

Boundary Condition for Magnetic field.



$$\oint \vec{H} \cdot d\vec{l} \quad \text{Ampere's law}$$



$$H_{t1} \Delta x + \cancel{H_n \Delta y} - H_{t2} \Delta x - \cancel{H_n \Delta y} = I$$

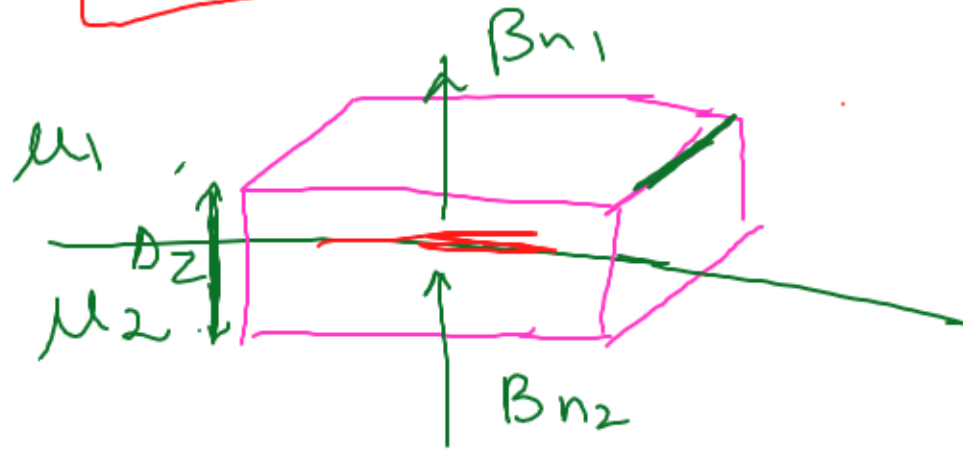
$$H_{t1} \Delta x - H_{t2} \Delta x = I$$

$$H_{t1} - H_{t2} = \frac{I}{\Delta x} = K$$



$$H_{t1} - H_{t2} = K$$

linear current density.



$$B_{n1}A - B_{n2}A = 0$$

$$B_{n1} - B_{n2} = 0$$

normal components of a magnetic field at the interface is continuous.



E field cannot lie tangent to the interface between media.



H field can lie tangent

Boundary conditions

E

H

① Tangential component always.

② Tangential component will be continuous dielectric
discontinuous dielectric
metal.

E

normal component
will be continuous
dielectric - dielectric
media

& discontinuous when
dielectric - metal.

H

normal components
will be continuous.

Power, $P = \int \mathbf{E} \cdot \mathbf{I}$

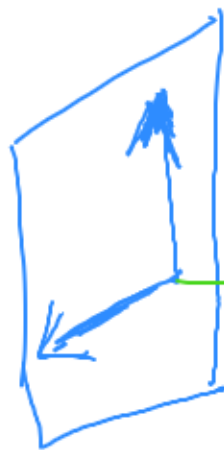
Power

$$P = \mathbf{E} \times \mathbf{H}$$

V/m

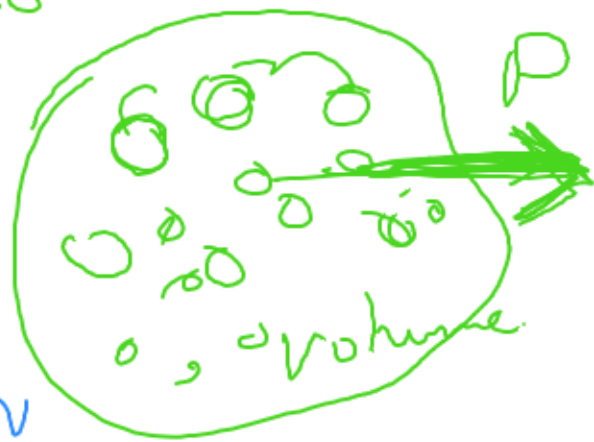
A/m

$\mathbf{E}$



$\mathbf{P}$

$$\int \nabla \cdot (\underline{\bar{E}} \times \underline{\bar{H}}) dv = \oint_S (\underline{\bar{E}} \times \underline{\bar{H}}) \cdot d\vec{s}$$



$$\int_{Vol.} (\underline{\bar{E}} \cdot \underline{\bar{J}}) dv = -\frac{\partial}{\partial t} \int_{Vol.} \left(\frac{\mu}{2} H^2 + \frac{\epsilon}{2} E^2 \right) dv - \oint_S (\underline{\bar{E}} \times \underline{\bar{H}}) \cdot d\vec{s}$$

The term $\int_V (\vec{E} \cdot \vec{J}) dV$ represents power dissipated
in the volume V

The term $-\frac{\partial}{\partial t} \int_V \left(\frac{\mu}{2} H^2 + \frac{\epsilon}{2} E^2 \right) dV$ represents the
stored electric and magnetic energy in the volume.

The term $-\oint (\vec{E} \times \vec{H}) \cdot d\vec{s}$ represents the rate of
flow of energy inward through the surface of the
volume.

$\vec{P} = \vec{E} \times \vec{H}$ is the Poynting vector used to measure the rate of energy flow per unit area at a point.

The direction of Poynting vector indicates the direction of the instantaneous power flow.

Radiation pattern

The radiation pattern is a graphical representation of the radiation (far field) properties of an antenna.

Isotropic antenna

An antenna which radiates equally in all directions. It is a point source



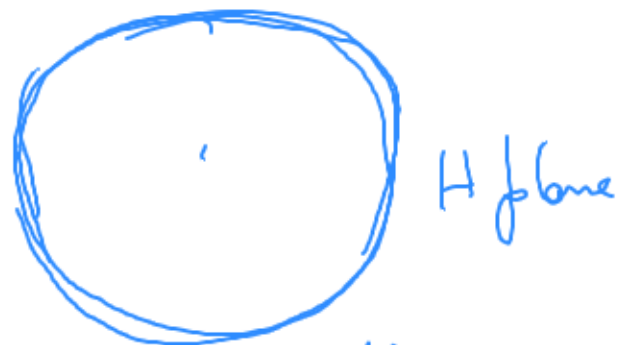
Directional antenna more

An antenna that radiates in a particular direction and less in other direction is called directional antenna



Omnidirectional antenna

It is an antenna that essentially has a non directional pattern in a given plane and directional pattern in orthogonal plane eg. dipole.



E plane



Principal pattern

The E and H plane are the principal planes of antenna pattern.

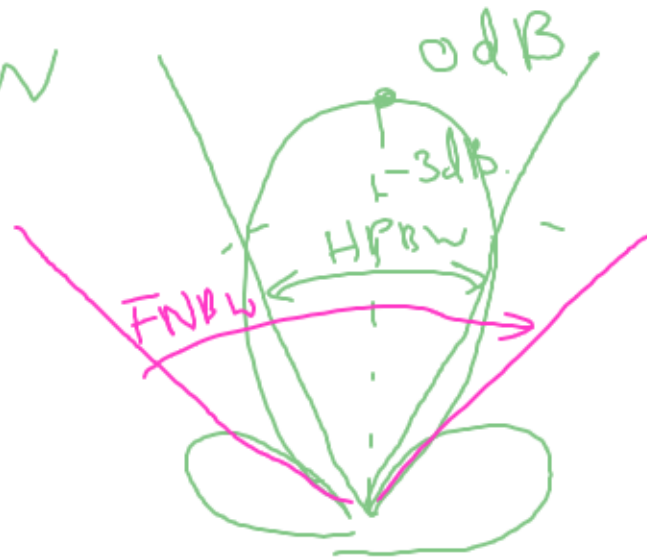
E plane :— It is a plane containing the electric field and the direction of maximum radiation

H plane - It is a plane containing magnetic field vector and the direction of maximum radiation

eg.



HPBW



The Beam width of the major lobe that spreads 3dB below the maximum value gives the angular measurement of HPBW.

Normalised power pattern

HPBW defined as the angular distance between the two points on the antenna pattern main lobe that are 3dB below the maximum gain plot.

$$HPBW = |\theta_{HPleft} - \theta_{HPright}|$$

FN BW (First null beamwidth) is defined as the angular distance between two points on each side of the major lobe (just adjacent) where the radiation drops to zero.

Revision

explained
8)

Gradient of V (characteristics)
 V is negative direction of $\vec{E}$ field.

9) Continuity equation

10) R Pattern, Principal pattern,

11) All done today.

12) Significance of Poynting vector.

All problems done in class practice.